

Lab2 : docker

Problem 1

Run a container nginx with name my-nginx and attach a and attach a volume 2 volumes to the container

- Volume1 for containing static html file
- Volume2 for containing nginx configuration

```
docker volume create html-volume
docker volume create conf-volume
docker run -d --name my-nginx -v html-volume:/usr/share/nginx/html -v conf-volume:/etc/nginx/conf.d -p 8080:80 nginx
curl localhost:8080
```

```
nagy@ZenBook-UX425EA:~/Mine/ITI/016_Docker/Solutions/Lab2$
nagy@ZenBook-UX425EA:~/Mine/ITI/016_Docker/Solutions/Lab2$ docker volume create html-volume
html-volume
nagy@ZenBook-UX425EA:~/Mine/ITI/016_Docker/Solutions/Lab2$ docker volume create conf-volume
conf-volume
nagy@ZenBook-UX425EA:~/Mine/ITI/016_Docker/Solutions/Lab2$
nagy@ZenBook-UX425EA:~/Mine/ITI/016_Docker/Solutions/Lab2$ docker run -d --name my-nginx -v html-volume:/usr/share/nginx/
html -v conf-volume:/etc/nginx/conf.d -p 8080:80 nginx
04818ab545a782d2e61fdc6f100482030d7782dbdab8238326664fa323b38148
nagy@ZenBook-UX425EA:~/Mine/ITI/016_Docker/Solutions/Lab2$
nagy@ZenBook-UX425EA:~/Mine/ITI/016_Docker/Solutions/Lab2$ curl localhost:8080
<!DOCTYPE html>
<html>
<head>
<title>Welcome to nginx!</title>
<style>
html { color-scheme: light dark; }
body { width: 35em; margin: 0 auto;
font-family: Tahoma, Verdana, Arial, sans-serif; }
</style>
</head>
<body>
<h1>Welcome to nginx!</h1>
<p>If you see this page, the nginx web server is successfully installed and
working. Further configuration is required.</p>

<p>For online documentation and support please refer to
<a href="http://nginx.org/">nginx.org</a>.<br/>
Commercial support is available at
<a href="http://nginx.com/">nginx.com</a>.</p>

<p><em>Thank you for using nginx.</em></p>
</body>
</html>
nagy@ZenBook-UX425EA:~/Mine/ITI/016_Docker/Solutions/Lab2$
nagy@ZenBook-UX425EA:~/Mine/ITI/016_Docker/Solutions/Lab2$
```

Edit the html content

```
docker exec -it my-nginx bash
echo "<h1> Hellooooooooo Woorldddd</h1>" > /usr/share/nginx/html/index.html
curl localhost:8080
```

```
nagy@ZenBook-UX425EA:~/Mine/ITI/016_Docker/Solutions/Lab2$
nagy@ZenBook-UX425EA:~/Mine/ITI/016_Docker/Solutions/Lab2$
nagy@ZenBook-UX425EA:~/Mine/ITI/016_Docker/Solutions/Lab2$ docker exec -it my-nginx bash
root@04818ab545a7:/#
root@04818ab545a7:/# echo "<h1> Hellooooooooo Woorlddddd</h1>" > /usr/share/nginx/html/index.html
root@04818ab545a7:/#
root@04818ab545a7:/# curl localhost:80
<h1> Hellooooooooo Woorlddddd</h1>
root@04818ab545a7:/#
root@04818ab545a7:/# exit
exit
nagy@ZenBook-UX425EA:~/Mine/ITI/016_Docker/Solutions/Lab2$
nagy@ZenBook-UX425EA:~/Mine/ITI/016_Docker/Solutions/Lab2$ curl localhost:8080
<h1> Hellooooooooo Woorlddddd</h1>
nagy@ZenBook-UX425EA:~/Mine/ITI/016_Docker/Solutions/Lab2$
nagy@ZenBook-UX425EA:~/Mine/ITI/016_Docker/Solutions/Lab2$
```

Remove the container

```
docker stop my-nginx
docker rm my-nginx
```

```
nagy@ZenBook-UX425EA:~/Mine/ITI/016_Docker/Solutions/Lab2$
nagy@ZenBook-UX425EA:~/Mine/ITI/016_Docker/Solutions/Lab2$
nagy@ZenBook-UX425EA:~/Mine/ITI/016_Docker/Solutions/Lab2$ docker stop my-nginx
my-nginx
nagy@ZenBook-UX425EA:~/Mine/ITI/016_Docker/Solutions/Lab2$
nagy@ZenBook-UX425EA:~/Mine/ITI/016_Docker/Solutions/Lab2$ docker rm my-nginx
my-nginx
nagy@ZenBook-UX425EA:~/Mine/ITI/016_Docker/Solutions/Lab2$
nagy@ZenBook-UX425EA:~/Mine/ITI/016_Docker/Solutions/Lab2$
```

Run a new 2 containers with the following:

- Attach the 2 volumes that was attached to the previous container in two different ways (volume mount bind mount)
- Map port 80 to port 8080 on you host machine
- Access the html files from your browser

```
docker run -d --name nginx-vol -v html-volume:/usr/share/nginx/html -v
conf-volume:/etc/nginx/conf.d -p 8080:80 nginx
docker run -d --name nginx-bind -v $(pwd)/html:/usr/share/nginx/html -v
$(pwd)/conf:/etc/nginx/conf.d -p 8088:80 nginx
curl localhost:8080
```

```
nagy@ZenBook-UX425EA:~/Mine/ITI/016_Docker/Solutions/Lab2$
nagy@ZenBook-UX425EA:~/Mine/ITI/016_Docker/Solutions/Lab2$
nagy@ZenBook-UX425EA:~/Mine/ITI/016_Docker/Solutions/Lab2$ docker run -d --name nginx-vol -v html-volume:/usr/share/nginx
/html -v conf-volume:/etc/nginx/conf.d -p 8080:80 nginx
dcf22aabe49a67eb4fa1560dbf79e6ecceaf401ca6050f2577b47d9e2331e2eb
nagy@ZenBook-UX425EA:~/Mine/ITI/016_Docker/Solutions/Lab2$
nagy@ZenBook-UX425EA:~/Mine/ITI/016_Docker/Solutions/Lab2$ docker run -d --name nginx-bind -v $(pwd)/html:/usr/share/nginx
/html -v $(pwd)/conf:/etc/nginx/conf.d -p 8088:80 nginx
79cd56f1033fab188ef5ella91789127346a86cb9e85b9ef9fccc7dff7d833df
nagy@ZenBook-UX425EA:~/Mine/ITI/016_Docker/Solutions/Lab2$
nagy@ZenBook-UX425EA:~/Mine/ITI/016_Docker/Solutions/Lab2$
nagy@ZenBook-UX425EA:~/Mine/ITI/016_Docker/Solutions/Lab2$
nagy@ZenBook-UX425EA:~/Mine/ITI/016_Docker/Solutions/Lab2$ curl localhost:8080
<h1> Hellooooooooo Woorlddddd</h1>
nagy@ZenBook-UX425EA:~/Mine/ITI/016_Docker/Solutions/Lab2$
nagy@ZenBook-UX425EA:~/Mine/ITI/016_Docker/Solutions/Lab2$
```

Problem 2

Create a dockerfile for nginx image with different html content and different nginx conf that listen to port 8080 instead of port 80 on the container

```
vim Dockerfile
vim nginx.conf
vim index.html
docker build -t my-nginx .
curl localhost:8080
```

```
FROM nginx

COPY index.html /usr/share/nginx/html/index.html
COPY nginx.conf /etc/nginx/conf.d/default.conf

EXPOSE 8080
```

```
nagy@ZenBook-UX425EA:~/Mine/ITI/016_Docker/Solutions/Lab2$
nagy@ZenBook-UX425EA:~/Mine/ITI/016_Docker/Solutions/Lab2$
nagy@ZenBook-UX425EA:~/Mine/ITI/016_Docker/Solutions/Lab2$ vim Dockerfile
nagy@ZenBook-UX425EA:~/Mine/ITI/016_Docker/Solutions/Lab2$
nagy@ZenBook-UX425EA:~/Mine/ITI/016_Docker/Solutions/Lab2$ vim nginx.conf
nagy@ZenBook-UX425EA:~/Mine/ITI/016_Docker/Solutions/Lab2$
nagy@ZenBook-UX425EA:~/Mine/ITI/016_Docker/Solutions/Lab2$ vim index.html
nagy@ZenBook-UX425EA:~/Mine/ITI/016_Docker/Solutions/Lab2$
nagy@ZenBook-UX425EA:~/Mine/ITI/016_Docker/Solutions/Lab2$
nagy@ZenBook-UX425EA:~/Mine/ITI/016_Docker/Solutions/Lab2$ docker build -t my-nginx .
[+] Building 0.3s (8/8) FINISHED                                docker:default
=> [internal] load build definition from Dockerfile              0.0s
=> => transferring dockerfile: 162B                             0.0s
=> [internal] load metadata for docker.io/library/nginx:latest  0.0s
=> [internal] load .dockerignore                                0.0s
=> => transferring context: 2B                                    0.0s
=> [1/3] FROM docker.io/library/nginx:latest                    0.1s
=> [internal] load build context                                0.0s
=> => transferring context: 230B                                  0.0s
=> [2/3] COPY index.html /usr/share/nginx/html/index.html      0.0s
=> [3/3] COPY nginx.conf /etc/nginx/conf.d/default.conf         0.0s
=> exporting to image                                           0.0s
=> => exporting layers                                           0.0s
=> => writing image sha256:ea26e9b9f9ce2d04d59e6cbdf48aa419b23750687136371d6e90192d9d6b5ff97 0.0s
=> => naming to docker.io/library/my-nginx                      0.0s
nagy@ZenBook-UX425EA:~/Mine/ITI/016_Docker/Solutions/Lab2$
nagy@ZenBook-UX425EA:~/Mine/ITI/016_Docker/Solutions/Lab2$
nagy@ZenBook-UX425EA:~/Mine/ITI/016_Docker/Solutions/Lab2$
nagy@ZenBook-UX425EA:~/Mine/ITI/016_Docker/Solutions/Lab2$ curl localhost:8080
<h1> Helloooooooooo Woorlddddd</h1>
nagy@ZenBook-UX425EA:~/Mine/ITI/016_Docker/Solutions/Lab2$
nagy@ZenBook-UX425EA:~/Mine/ITI/016_Docker/Solutions/Lab2$
```

Problem 3

Create a reactjs simple app

```
npx create-react-app demo
cd demo
```

Create a dockerfile to containerize the reactapp

```
vim Dockerfile
```

```
# Stage 1: Build React app
FROM node:18 AS build
WORKDIR /app
COPY package*.json ./
RUN npm install
COPY . .
RUN npm run build

# Stage 2: Serve with nginx
FROM nginx:alpine
COPY --from=build /app/build /usr/share/nginx/html
EXPOSE 80
CMD ["nginx", "-g", "daemon off;"]
```

Build the image and test it

```
docker build -t my-app .
docker run -d -p 3000:80 --name react-app my-app
curl localhost:3000
```

```
nagy@ZenBook-UX425EA:~/Mine/ITI/016_Docker/Solutions/Lab2/demo$ docker build -t my-app .
[+] Building 17.3s (14/14) FINISHED
=> [internal] load build definition from Dockerfile 0.0s
=> => transferring dockerfile: 308B 0.0s
=> [internal] load metadata for docker.io/library/nginx:alpine 0.6s
=> [internal] load metadata for docker.io/library/node:18 0.6s
=> [internal] load .dockerignore 0.0s
=> => transferring context: 2B 0.0s
=> [build 1/6] FROM docker.io/library/node:18@sha256:c6ae79e38498325db67193d391e6ecd224d96c693a8a4d943498556716d 0.0s
=> [stage-1 1/2] FROM docker.io/library/nginx:alpine@sha256:1d13701a5f9f3fb01aaa88cef2344d65b6b5bf6b7d9fa4cf0dca5 0.0s
=> [internal] load build context 1.0s
=> => transferring context: 3.03MB 0.9s
=> CACHED [build 2/6] WORKDIR /app 0.0s
=> CACHED [build 3/6] COPY package*.json ./ 0.0s
=> CACHED [build 4/6] RUN npm install 0.0s
=> [build 5/6] COPY . . 8.6s
=> [build 6/6] RUN npm run build 6.6s
=> CACHED [stage-1 2/2] COPY --from=build /app/build /usr/share/nginx/html 0.0s
=> exporting to image 0.0s
=> => exporting layers 0.0s
=> => writing image sha256:ae3200b19f2787d1d27caf2c40785b92698c4069a363d856f33331d87c53fa6e 0.0s
=> => naming to docker.io/library/my-app 0.0s
nagy@ZenBook-UX425EA:~/Mine/ITI/016_Docker/Solutions/Lab2/demo$
nagy@ZenBook-UX425EA:~/Mine/ITI/016_Docker/Solutions/Lab2/demo$
nagy@ZenBook-UX425EA:~/Mine/ITI/016_Docker/Solutions/Lab2/demo$ docker run -d -p 3000:80 --name react-app my-app
9d7b5ca52bd4de0d71a68a105bc7636d345e9934289649402f58159c9f287547
nagy@ZenBook-UX425EA:~/Mine/ITI/016_Docker/Solutions/Lab2/demo$
nagy@ZenBook-UX425EA:~/Mine/ITI/016_Docker/Solutions/Lab2/demo$ docker ps
CONTAINER ID   IMAGE          COMMAND                  CREATED        STATUS        PORTS
NAMES
9d7b5ca52bd4   my-app        "/docker-entrypoint.    6 seconds ago Up 5 seconds   0.0.0.0:3000->80/tcp, [::]:3000->80
/tcp          react-app
878d1acdcl12   mysql:latest  "docker-entrypoint.s    16 hours ago  Up 16 hours   3306/tcp, 33060/tcp
app-database
nagy@ZenBook-UX425EA:~/Mine/ITI/016_Docker/Solutions/Lab2/demo$
nagy@ZenBook-UX425EA:~/Mine/ITI/016_Docker/Solutions/Lab2/demo$ curl localhost:3000
<!doctype html><html lang="en"><head><meta charset="utf-8"/><link rel="icon" href="/favicon.ico"/><meta name="viewport" c
ontent="width=device-width,initial-scale=1"/><meta name="theme-color" content="#000000"/><meta name="description" content
="Web site created using create-react-app"/><link rel="apple-touch-icon" href="/logo192.png"/><link rel="manifest" href="
/manifest.json"/><title>React App</title><script defer="defer" src="/static/js/main.dab249f8.js"></script><link href="/st
atic/css/main.f855e6bc.css" rel="stylesheet"/></head><body><noscript>You need to enable JavaScript to run this app.</noscr
ipt><div id="root"></div></body></html>nagy@ZenBook-UX425EA:~/Mine/ITI/016_Docker/Solutions/Lab2/demo$
nagy@ZenBook-UX425EA:~/Mine/ITI/016_Docker/Solutions/Lab2/demo$
```

